

Project Title: Progress Toward Health-Related Sustainable Development Goals.

Progress Toward Health-Related Sustainable Development Goals Assessing Universal Health Coverage Progress (SDG 3.8)

Executive Summary

Universal Health Coverage (UHC) is a central part of Sustainable Development Goal 3.8 because it reflects whether people can access essential health services without suffering financial hardship when using them (World Health Organization, 2025; United Nations, 2015). As 2030 gets closer, the key policy question is no longer whether UHC matters, but whether current progress is strong, broad, and resilient enough across countries to move meaningfully toward the target (United Nations, 2015; World Health Organization and World Bank, 2025).

This report assesses progress toward UHC between 2000 and 2022, focusing on variation across world regions and World Bank income groups. The analysis uses three linked indicators: the UHC service coverage index (SDG 3.8.1), financial protection through catastrophic health expenditure (SDG 3.8.2), and health expenditure per capita (United Nations Statistics Division, 2025a; United Nations Statistics Division, 2025b; World Bank, 2022). These indicators were selected because they provide a broader picture of UHC than service access alone. Together, they capture service availability, affordability, and the level of resource commitment within national health systems.

The report is based on secondary data drawn from internationally recognised sources, including the United Nations, the World Health Organization, the World Bank, and Our World in Data (World Health Organization, 2023a; World Bank, 2022; Our World in Data, 2026). The analysis period was restricted to 2000–2022 because 2022 represented the most recent year for which the key indicators were sufficiently comparable across countries and datasets. Although later observations were available for some variables, they were more uneven across countries and indicators, which would have weakened cross-country consistency.

The findings show that progress toward UHC has occurred, but it has not been evenly distributed. Global service coverage improved over time, indicating that many countries expanded access to essential health services during the period. However, the pace and depth of progress varied substantially. Higher-income countries generally showed stronger average service coverage across the period, while lower-income groups remained more exposed to weaker outcomes and slower long-term improvement. Regional comparison shows a similar pattern, with some regions maintaining higher average coverage levels while others remained comparatively lower across much of the period.

The analysis also identifies a positive relationship between health expenditure and UHC service coverage, but this relationship is not simple or automatic. Countries that spend more

on health often achieve stronger service coverage, yet higher expenditure does not by itself guarantee stronger outcomes. This suggests that the effectiveness of spending matters alongside the level of spending itself (World Bank, 2022; World Health Organization and World Bank, 2025; Kruk et al., 2018). Financial protection adds an important complementary dimension by showing that service expansion alone is not enough if households remain exposed to severe out-of-pocket costs (United Nations Statistics Division, 2025b; Wagstaff et al., 2018).

The report also considers the COVID-19 period and indicative progress toward 2030. Recent disruption highlighted differences in resilience, with some countries and groups appearing better able to retain or recover trajectory under pressure than others (World Health Organization and World Bank, 2025; Kruk et al., 2018). A simple projection based on long-term historical change suggests that future trajectories are also uneven. These estimates are indicative rather than predictive, but they help identify where progress appears comparatively stronger and where the risk of remaining behind remains greatest.

Overall, the report concludes that UHC progress is advancing, but through unequal and structurally shaped pathways rather than through a single shared global pattern. For decision-makers, this means policy attention, financial protection measures, and health-system support should be targeted more sharply toward countries and groups where coverage remains weaker and projected momentum toward 2030 appears less robust.

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1. Background and Problem Definition

Healthcare is central to human development because it affects health outcomes, poverty reduction, labour productivity, and long-term development. Progress remains incomplete if large parts of the population cannot access care or face serious financial pressure when doing so (United Nations, 2015; World Bank, 2023). Health therefore occupies a prominent place within the Sustainable Development Goals, particularly SDG 3 (United Nations, 2015).

Within SDG 3, Target 3.8 focuses specifically on Universal Health Coverage (UHC). UHC is generally understood as ensuring that all people can access the health services they need without suffering financial hardship as a result (World Health Organization, 2025). This definition shows that UHC is not a single-dimensional concept. It depends on both service access and affordability, since services are insufficient if they remain financially inaccessible, and financial protection alone is insufficient if essential services remain weak or unevenly distributed (World Health Organization and World Bank, 2025).

This report examines UHC through three interconnected dimensions. The first is UHC service coverage, which provides a comparative indication of access to essential health services. The second is financial protection, which reflects whether households are exposed to catastrophic health spending when seeking care. The third is health expenditure per capita, which is used here as an indicator of the level of resource commitment within a country's health system (United Nations Statistics Division, 2025a; United Nations Statistics Division, 2025b; World Bank, 2022). Considering these dimensions together creates a more complete picture of progress than relying on service access alone.

The main issue explored in this report is not just whether progress toward UHC has happened, but how that progress differs across countries, regions, and income groups, and what those differences mean for future decision-making. Global averages can create the impression of broad improvement while masking deep inequality in performance. If some countries are making relatively strong gains while others continue to lag, then a general global narrative of improvement may overstate how close the world really is to achieving UHC in practice (World Health Organization and World Bank, 2025; World Bank, 2023).

This issue becomes more urgent as the 2030 SDG deadline approaches (United Nations, 2015). The current policy question is not only whether countries have improved, but whether observed trajectories are strong enough to support meaningful progress by 2030. This is not simply a descriptive question. It is also a decision-support problem. Governments, development agencies, and international health institutions operate under fiscal and institutional constraints. If variation in progress is not understood clearly, resources may be allocated too broadly, too weakly, or without sufficient focus on the countries and groups facing the most serious barriers.

Because this report focuses specifically on SDG 3.8 rather than all SDG 3 targets, the question of what is on track or lagging is addressed here through UHC progress, resilience, and indicative 2030 trajectory.

The COVID-19 pandemic sharpened this challenge. The disruption caused by the pandemic did not affect all health systems equally. Some countries appear to have retained service continuity more effectively, while others experienced visible disruption or weaker recovery (World Health Organization and World Bank, 2025; Our World in Data, 2026). This means that UHC progress now needs to be assessed not only in terms of long-term improvement, but also in terms of resilience. A health system that improves only under stable conditions may still be vulnerable if it cannot sustain progress under pressure.

From a Business Analysis and Consulting perspective, this project can be understood as a problem of comparative performance, prioritisation, and evidence-based decision support. The aim is not only to describe international health indicators, but to turn them into a clearer assessment of where progress is strongest, where it is weakest, and what practical lessons can

be drawn from that variation. The value of the analysis lies in helping decision-makers look beyond global averages and develop a clearer basis for prioritisation and support.

The report therefore remains tightly focused on SDG 3.8 so that the analysis stays specific, comparable, and directly aligned with the project brief. The report is structured around four main questions. First, how has UHC service coverage changed over time? Second, how does progress differ across regions and income groups? Third, is health expenditure associated with stronger service coverage outcomes? Fourth, what does recent performance, including the COVID-disruption period, suggest about resilience and likely future trajectory?

Taken together, these questions move the report beyond simple monitoring toward a more decision-relevant assessment of comparative UHC progress.

2. Literature Review

Universal Health Coverage has become one of the most important ideas in contemporary global health policy because it links equity, access, and financial protection within a single framework (World Health Organization, 2025; United Nations, 2015). At its core, UHC reflects the principle that health systems should provide essential services to all people without exposing them to severe financial burden. International organisations strongly support this principle, but the literature makes clear that UHC is not a simple end state. Rather, it is a gradual and uneven process shaped by financing structures, administrative capacity, political priorities, and broader socio-economic conditions.

A major theme in the literature is that UHC should be understood through at least two central dimensions: coverage of essential services and protection from financial hardship (World Health Organization and World Bank, 2025; United Nations Statistics Division, 2025a; United Nations Statistics Division, 2025b). This distinction matters because countries may perform relatively well on one dimension while remaining weak on the other, so a full assessment of UHC requires more than a single access indicator.

The measurement of SDG 3.8 reflects this multidimensional logic. The UHC service coverage index, reported under SDG 3.8.1, is widely used because it provides a composite measure of coverage across a set of tracer service areas. These include reproductive, maternal, newborn and child health, infectious diseases, non-communicable diseases, and service capacity and access (United Nations Statistics Division, 2025a; World Health Organization, 2023a). This makes the index useful for long-term comparison across countries and over time. However, the literature also recognises that a composite index cannot fully capture all aspects of health-system performance, particularly where quality, internal inequalities, or subnational disparities vary significantly.

For this reason, financial protection remains a critical complementary dimension. The literature on catastrophic health expenditure shows that access to services cannot be treated as a sufficient condition for UHC if using those services still creates major economic strain for households (United Nations Statistics Division, 2025b; Wagstaff et al., 2018). Financial hardship can discourage care-seeking, delay treatment, and deepen poverty, especially where out-of-pocket payment remains a major barrier. The literature therefore supports the idea that access and affordability are inseparable within any meaningful understanding of UHC.

Another important area of the literature concerns inequality across countries, regions, and income groups. Progress toward UHC is repeatedly described as uneven rather than uniform. Higher-income countries generally tend to record stronger service coverage because they have broader fiscal capacity, stronger institutions, and more established health infrastructure. By contrast, lower-income countries often face constraints related to financing, workforce shortages, service delivery limitations, and weaker administrative systems (World Health Organization and World Bank, 2025; World Bank, 2023). This means that variation in UHC outcomes is not random, but is connected to wider structural and economic conditions.

Health financing is also a major theme in the literature. Many studies and global monitoring reports identify a positive relationship between health expenditure and UHC outcomes (World Bank, 2022; World Health Organization and World Bank, 2025). Countries that spend more on health frequently achieve stronger service coverage and more robust system capacity. However, the literature also warns against assuming that more spending always leads directly to better outcomes. The efficiency, targeting, and governance of expenditure matter greatly. Some countries appear able to translate moderate levels of spending into relatively strong progress, while others do not convert higher expenditure into equivalent improvements. This distinction between how much is spent and how effectively it is used matters for policy because it shifts attention away from inputs alone and toward system performance and institutional capability (Kruk et al., 2018).

The literature also increasingly emphasises health-system resilience. The COVID-19 pandemic exposed how quickly health-service progress can be disrupted by external shocks. Some systems that had been improving over time were less able to maintain essential services under pressure, while others retained continuity more effectively (Kruk et al., 2018; World Health Organization and World Bank, 2025). This has shifted attention from simple expansion of coverage toward the resilience of that expansion. Progress is now understood not only in terms of growth over time, but also in terms of durability under stress.

A further theme concerns future trajectory. As 2030 approaches, global monitoring has moved from broad ambition toward a more operational question: whether current rates of change appear sufficient to support meaningful progress by the SDG deadline (United Nations, 2015; World Health Organization and World Bank, 2025). This has encouraged the use of trajectory analysis, indicative projection, and gap-based interpretation. Such approaches are useful because they translate past and present patterns into a forward-looking assessment, while still distinguishing between indicative projection and definitive prediction. Historical trends can suggest direction, but future outcomes remain vulnerable to policy change, fiscal constraints, demographic pressures, and external shocks.

Taken together, the literature suggests four key points relevant to this report. First, UHC must be understood through both service coverage and financial protection. Second, inequality across countries, regions, and income groups is central to understanding progress. Third, expenditure matters, but the effectiveness of spending matters as well. Fourth, resilience and future trajectory are increasingly important because the key policy question is no longer only whether progress has happened, but whether it appears strong enough to continue toward 2030.

This report builds on that literature by using internationally comparable indicators to produce a client-facing comparative assessment of UHC progress. It does not try to explain every national difference in a fully causal way. Instead, it focuses on showing where progress has

been stronger or weaker, how that variation appears across groups, and what those patterns imply for future prioritisation and support.

3. Methodology

This report uses a quantitative, comparative, secondary-data approach to analyse progress toward Universal Health Coverage under SDG 3.8. The aim is to compare progress across countries, regions, and income groups using internationally comparable indicators, rather than to evaluate one national case in depth (United Nations, 2015; World Health Organization and World Bank, 2025). Secondary data were therefore appropriate because they allowed broad country coverage, long-term comparison, and alignment with the approved international data sources highlighted in the project brief and kickoff guidance.

The analysis is based on three core indicators that reflect the main dimensions of UHC. The first is SDG Indicator 3.8.1, the UHC service coverage index. This index provides a composite measure of coverage of essential services across a set of tracer indicators and serves as the main outcome variable in the report because it offers the clearest internationally comparable measure of access to essential health services over time (United Nations Statistics Division, 2025a; World Health Organization, 2023a).

The second indicator is SDG Indicator 3.8.2, financial protection. This captures the share of the population experiencing catastrophic health expenditure and is included because access to care alone does not provide a complete picture of UHC if households remain exposed to severe financial hardship when using healthcare (United Nations Statistics Division, 2025b; Wagstaff et al., 2018). It therefore functions as a complementary measure of whether service expansion is financially sustainable.

The third indicator is health expenditure per capita. This variable is used as a proxy for health-system investment and is included to explore whether higher levels of spending are associated with stronger service coverage outcomes (World Bank, 2022). In this report, expenditure is used analytically as an associational variable rather than as a direct causal explanation of performance.

The main data sources were the United Nations SDG Global Database, the World Health Organization Global Health Observatory, World Bank classification and development indicator data, and Our World in Data (United Nations Statistics Division, 2025a; United Nations Statistics Division, 2025b; World Health Organization, 2023a; World Bank, 2022; Our World in Data, 2026). These were selected because they provide internationally recognised indicators, broad country coverage, and sufficient consistency for long-term comparative analysis.

The analysis period was restricted to 2000–2022. This decision was taken because 2022 represented the most recent year for which the key indicators were relatively comparable across countries and datasets. Although later observations were available for some variables, they were more uneven across countries and indicators due to reporting lags and inconsistent coverage. Restricting the analysis to 2000–2022 therefore improved comparability and reduced the risk of misleading cross-country conclusions based on partial data (World Health Organization and World Bank, 2025).

Before analysis, the datasets were cleaned and standardised. Country names differed across sources, so a country-name harmonisation process was used to match source names with World Bank economy names. World Bank region and income-group classifications were then attached to the country records to support group-level comparison. The cleaned datasets were then merged into a single country-year analytical dataset containing country, harmonised World Bank country name, year, region, income group, UHC service coverage, financial protection, and health expenditure per capita.

Some missing values remained after the merge because not all indicators are reported annually for all countries. Rather than removing large portions of the data immediately, the merged dataset retained observations wherever possible and used filtering where necessary depending on the analytical task. In grouped regional and income analyses, this also required filtered subsets that excluded unclassified observations where source entities were not present in the World Bank classification file.

The analytical design consists of six main components. First, trend analysis was used to examine how UHC service coverage changed over time globally. This provides the broadest view of whether service coverage improved over the long term.

Second, comparative analysis by region and income group was used to identify structural differences in performance. This directly addresses variation across world regions and income groups and shows whether inequality in progress remains persistent.

Third, country-level summary analysis was used to identify stronger and weaker patterns of progress. This helps distinguish between countries with relatively strong momentum and those where long-term improvement has remained limited. To support this comparison, an average annual rate of change was calculated for each country using the difference between UHC service coverage in 2000 and 2022 divided by the number of years in the interval. This was expressed as:

$$\text{Annual Change 2000–2022} = (\text{UHC 2022} - \text{UHC 2000}) / 22$$

This measure was used to compare long-term momentum in UHC improvement across countries.

Fourth, a cross-sectional analysis of health expenditure and service coverage was used to examine whether higher levels of spending were associated with stronger UHC outcomes. This relationship is interpreted as associative rather than causal because the dataset does not include the broader set of institutional, demographic, and policy variables required to identify full causal drivers of performance (Kruk et al., 2018). It therefore provides a comparative indication of whether stronger UHC coverage tends to be associated with higher health-system spending, without making a causal claim.

Fifth, the report includes a simple forward-looking projection to support consideration of the 2030 objective. Projected UHC values for 2030 were calculated by extending each country's historical average annual rate of change from 2000–2022 forward by eight years. The calculation used was:

$$\text{Projected UHC 2030} = \text{UHC 2022} + (\text{Annual Change 2000–2022} \times 8)$$

This produces an indicative estimate of future trajectory if recent long-term patterns continue. It is intended as an analytical support tool rather than a precise forecast or official SDG attainment model (United Nations, 2015; World Health Organization and World Bank, 2025).

The methodology also includes a descriptive COVID-period perspective. To provide a simple indication of disruption during the pandemic period, recent change was examined using the difference between 2019 and 2021 UHC values:

$$\text{COVID-period change} = \text{UHC 2021} - \text{UHC 2019}$$

This was used as a descriptive proxy for disruption or resilience rather than as a formal estimate of pandemic impact.

Overall, the methodology is designed to balance breadth, comparability, and decision relevance. It provides a structured basis for identifying patterns, comparing performance, and assessing whether observed trajectories appear strong or weak as 2030 approaches.

4. Findings and Results

The analysis shows that progress toward Universal Health Coverage between 2000 and 2022 was real, but unevenly distributed. Service coverage improved over time, indicating that many countries expanded access to essential health services during the period. As shown in Figure 1, the overall global pattern is upward rather than flat, which supports the conclusion that progress has occurred over the long term. However, this broad upward trend should not be interpreted as evidence of a shared global pathway. The overall increase in coverage coexisted with persistent variation across regions, income groups, and individual country trajectories.

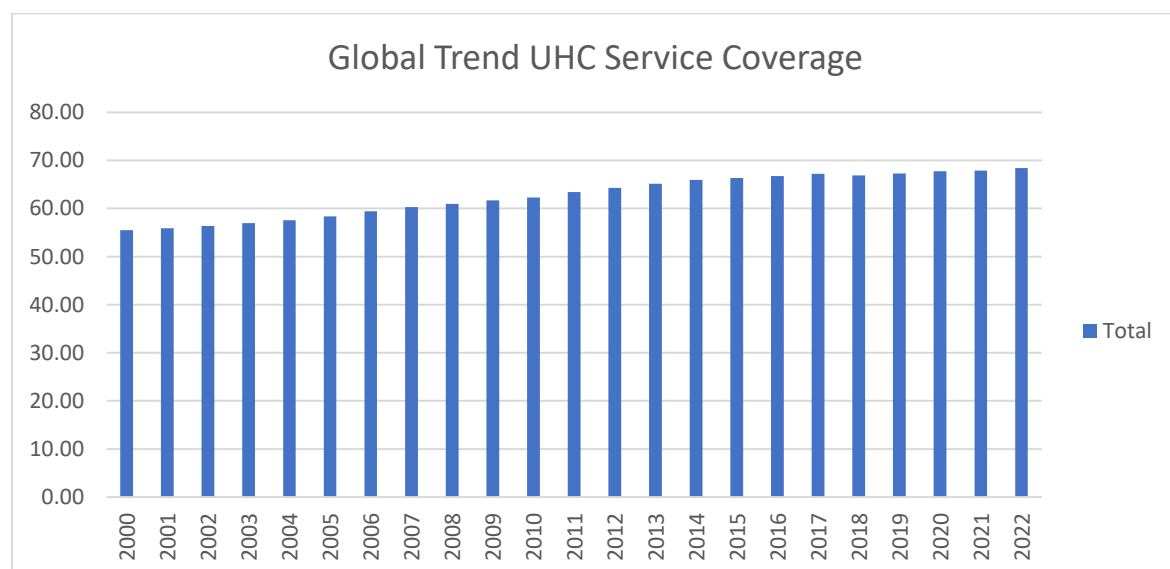


Figure 1. Global trend in UHC service coverage, 2000–2022

Regional comparison reinforces this point. Figure 2 shows that average UHC service coverage differed across regions throughout the period, with persistent separation between

higher-coverage and lower-coverage regional averages. This suggests that regional context continues to shape both the level and pace of UHC progress.

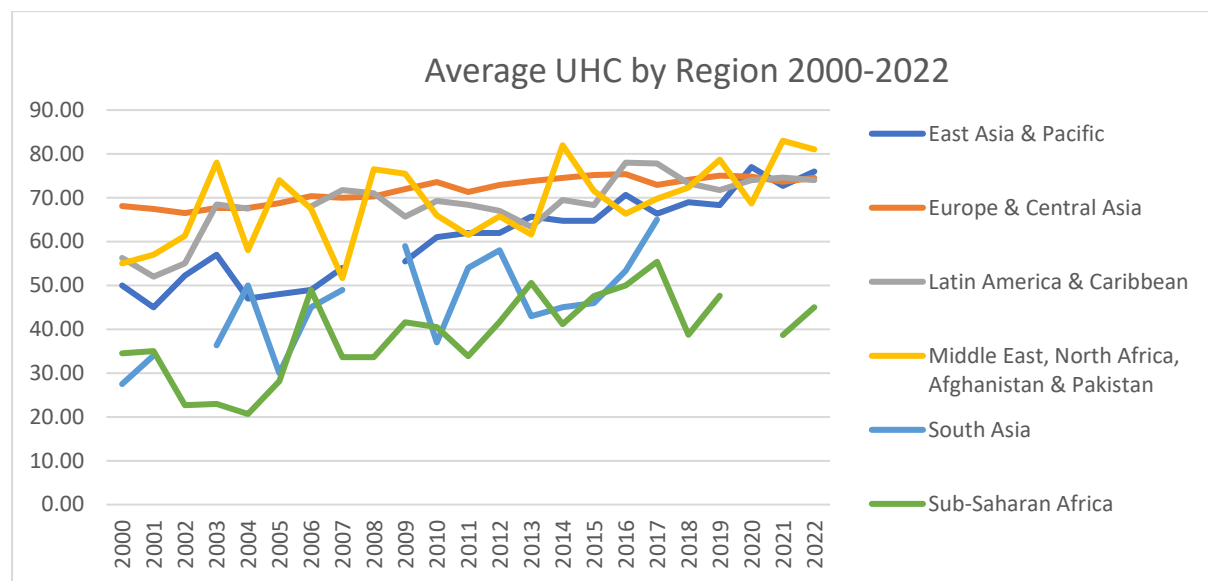


Figure 2. Average UHC service coverage by region, 2000–2022

A similar pattern emerges across income groups. Figure 3 indicates that high-income countries generally show stronger average service coverage, while lower-income groups remain more exposed to weaker outcomes and slower long-term improvement. This pattern is consistent with broad differences in fiscal space, administrative capacity, and long-term health-system investment. The finding does not imply that all countries within the same income group perform equally. However, it does indicate that economic position remains strongly associated with comparative UHC outcomes.

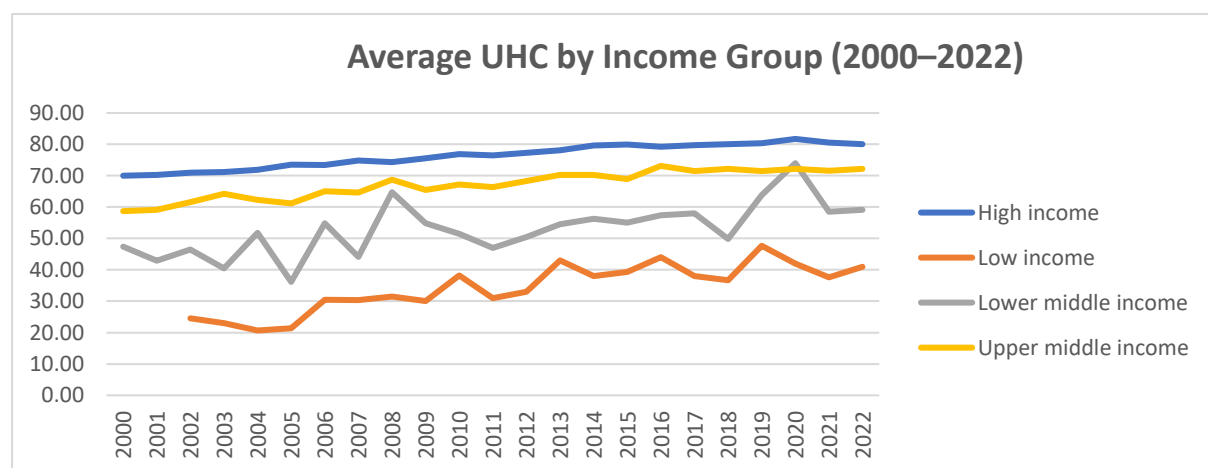


Figure 3. Average UHC service coverage by World Bank income group, 2000–2022

The country-level analysis adds an important layer to this picture. Table 1 shows selected countries with the strongest recorded improvement between 2000 and 2022, demonstrating that strong historical progress is not confined to one single region or income category. This shows that variation is not fully captured by region or income group alone, and that structural conditions are relevant but do not fully determine performance (World Health Organization

and World Bank, 2025; World Bank, 2023). Countries within the same broad category can still differ substantially in how effectively they convert investment and policy effort into improved outcomes.

Countries with stronger improvement trajectories are particularly important because they show that meaningful progress is possible even under constraint (World Health Organization and World Bank, 2025; Kruk et al., 2018). As shown in Table 1, several of the strongest improvers come from lower-income or lower-middle-income settings, indicating that strong long-term improvement is possible even where average structural constraints remain. This matters because slower long-term improvement raises concern about whether progress will be sufficient by 2030 if present patterns continue.

Country	Region	Income_Group	AnnualChange_2000_2022	Change_2000_2022
Rwanda	Sub-Saharan Africa	Low income	1.82	40
Nepal	South Asia	Lower middle income	1.73	38
India	South Asia	Lower middle income	1.68	37
Cambodia	East Asia & Pacific	Lower middle income	1.55	34
Bangladesh	South Asia	Lower middle income	1.36	30
Lao People's Democratic Republic	East Asia & Pacific	Lower middle income	1.32	29
Ghana	Sub-Saharan Africa	Lower middle income	1.32	29
Burkina Faso	Sub-Saharan Africa	Low income	1.32	29
Kenya	Sub-Saharan Africa	Lower middle income	1.27	28

Table 1. Selected countries with strongest improvement in UHC service coverage, 2000–2022

The table is presented as an illustrative comparison of stronger historical improvers rather than as proof of a single shared causal pathway. In relation to the project question of what distinguishes faster progress, the comparative evidence suggests that stronger improvers tend to combine higher long-term momentum, more effective translation of spending into service

gains, and, in some cases, better resilience during the COVID-period disruption. The analysis therefore suggests that faster progress is associated with resource commitment, spending effectiveness, and system resilience, while also showing that these relationships cannot be treated as fully causal within the limits of the available dataset (World Health Organization and World Bank, 2025; Kruk et al., 2018).

A short illustrative case comparison helps make this pattern more concrete. Rwanda and Nepal are useful examples because both appear among the stronger improvers despite lower-income or lower-middle-income structural conditions. While this report does not undertake a full qualitative case study of national policy systems, these countries act as indicative cases showing that comparatively strong progress is possible outside high-income settings. Their performance suggests that comparatively strong improvement can occur under constraint, even though the present dataset does not allow a full causal explanation of why these countries outperformed weaker peers.

The relationship between health expenditure and service coverage provides another key finding. The cross-sectional analysis suggests a positive but non-linear association between spending and coverage. Figure 4 also shows that countries do not convert spending into outcomes in identical ways. Some achieve comparatively strong outcomes at moderate spending levels, while others do not appear to translate higher expenditure into proportionate gains.

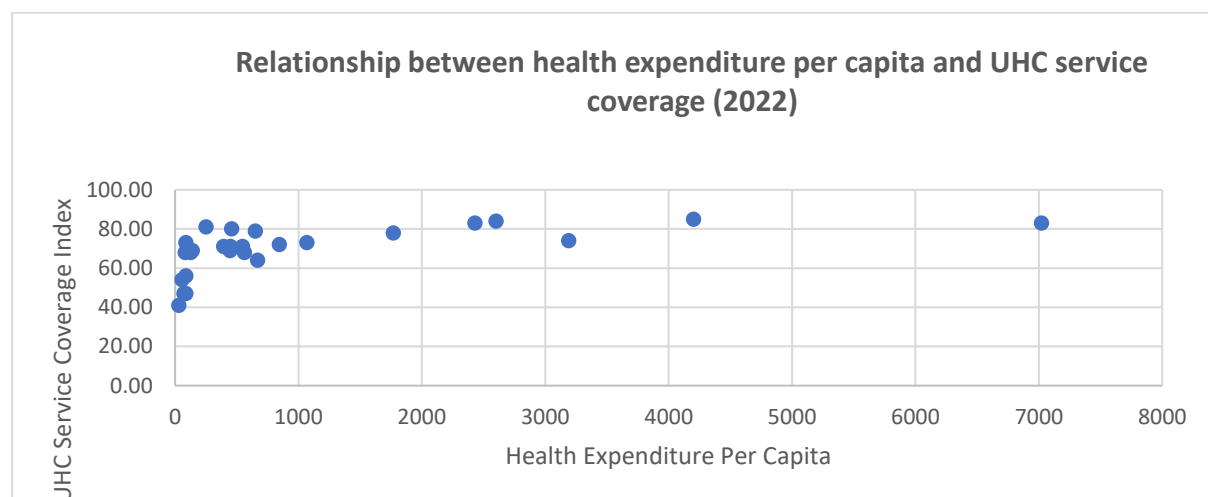


Figure 4. Health expenditure per capita and UHC service coverage, 2022

This suggests that the level of spending matters, but so does the effectiveness of spending. The observed pattern is consistent with the view that broader service expansion depends not only on resources, but also on how allocation, system organisation, governance, and implementation capacity shape whether expenditure produces stronger service coverage (World Bank, 2022; World Health Organization and World Bank, 2025; Kruk et al., 2018). From a decision-making perspective, this means that weaker-performing countries may require not only greater investment, but also better translation of investment into accessible and effective services.

Financial protection adds an important complementary dimension to the interpretation of the findings. The inclusion of the financial protection indicator reinforces that service access

alone is not sufficient to capture meaningful UHC progress if households remain exposed to catastrophic out-of-pocket costs. This suggests that service coverage may improve while affordability remains weak in some contexts. This is particularly important because UHC is defined not only by service availability, but also by the ability to use services without severe financial hardship (United Nations Statistics Division, 2025b; Wagstaff et al., 2018).

Although the financial protection analysis is more constrained by data availability than the service coverage analysis, it still supports an important conclusion: progress in UHC should be interpreted in multidimensional terms. Systems that improve service coverage without improving financial protection may still leave households vulnerable and may discourage timely care seeking.

This means the financial protection dimension is used primarily to strengthen interpretation of multidimensional UHC progress rather than to support the same depth of cross-country comparison as the service-coverage analysis (United Nations Statistics Division, 2025b; Wagstaff et al., 2018).

The COVID-period analysis provides an additional perspective on resilience and on how coverage trajectories changed under disruption. Comparing recent changes around 2019–2021 suggests that not all countries and groups maintained progress equally well under pressure. Figure 5 is used here as a descriptive comparison rather than a causal estimate, and it helps show whether recent UHC trajectories appear to have held, weakened, or recovered differently across regions during the COVID-period. Some appear to have retained or recovered recent progress more effectively, while others experienced more visible disruption. This suggests that health-system progress should be assessed not only by long-term upward movement, but also by the ability to absorb shock while maintaining core services.

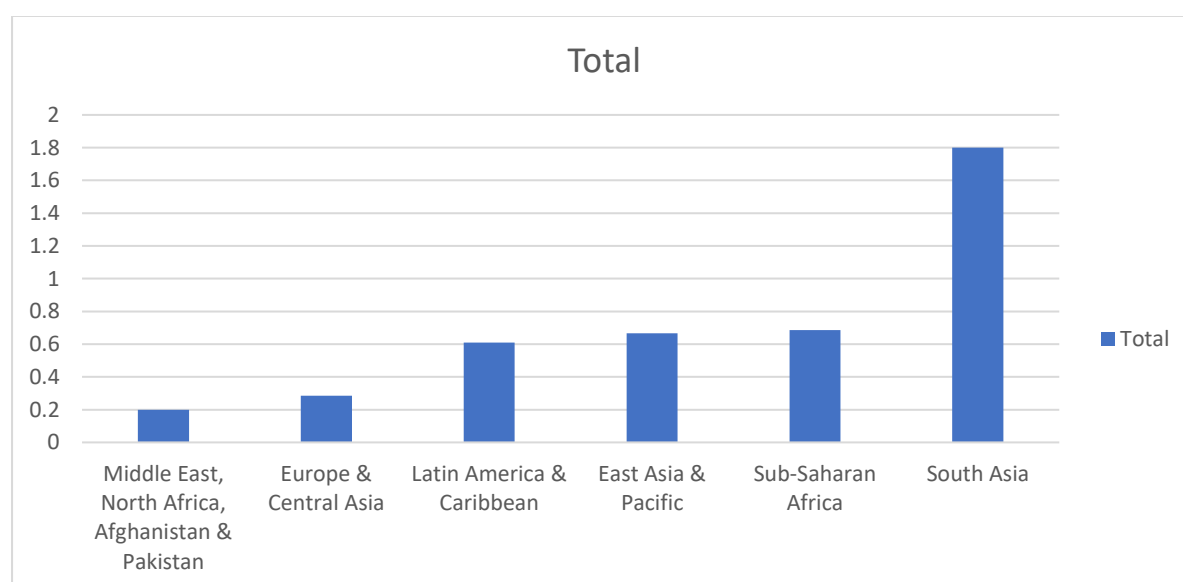


Figure 5. Descriptive COVID-period change in UHC service coverage by region, 2019–2021

Resilience therefore emerges as an important interpretive theme. Countries showing more durable progress are not only those that improve during stable periods, but also those that maintain continuity under pressure. The pandemic period made visible a distinction between systems that are progressing and systems that are progressing in a durable way. This has

practical significance because the path to 2030 is likely to remain shaped by uncertainty, fiscal pressure, and external shocks.

To address the project objective of considering progress toward 2030, the report also used a simple projection based on historical average annual change between 2000 and 2022. For each country, the projected 2030 value was estimated by adding eight years of average annual historical change to the 2022 UHC score. These projected values do not represent formal forecasts, but they are useful as indicative trajectory measures. They suggest that countries and groups with stronger historical momentum are more likely to occupy comparatively stronger positions by 2030 if recent patterns continue, while countries with weaker historical momentum may remain behind unless there is stronger policy change or additional support.

Table 2. Selected countries with strongest projected UHC trajectory to 2030

Country	Region	Income_Group	UHC_2022	AnnualChange_2000_2022	Projected_UHC_2030
Rwanda	Sub-Saharan Africa	Low income	58	1.82	72.5
Nepal	South Asia	Lower middle income	56	1.73	69.8
India	South Asia	Lower middle income	68	1.68	81.5
Cambodia	East Asia & Pacific	Lower middle income	60	1.55	72.4
Bangladesh	South Asia	Lower middle income	54	1.36	64.9
Lao People's Democratic Republic	East Asia & Pacific	Lower middle income	62	1.32	72.5

Ghana	Sub-Saharan Africa	Lower middle income	57	1.32	67.5
Burkina Faso	Sub-Saharan Africa	Low income	48	1.32	58.5
Kenya	Sub-Saharan Africa	Lower middle income	57	1.27	67.2

Table 2 adds a more visible forward-looking comparison to the report by showing which countries appear to have relatively stronger projected momentum if recent long-term patterns continue. These projected values should be interpreted as indicative rather than predictive, but they help show that future progress is unlikely to be evenly distributed.

This forward-looking comparison highlights that the likelihood of stronger future performance is unevenly distributed. Some countries and groups appear to be following relatively stronger projected trajectories, while others remain vulnerable to slower progress. This forward-looking dimension is particularly important because it connects historical performance with the policy problem of prioritisation. If recent trajectories are not equally strong, then the risk of remaining behind by 2030 is also not equally distributed (World Health Organization and World Bank, 2025; Kruk et al., 2018).

Taken together, the findings point to one clear overall conclusion: UHC progress is taking place, but not through a single global pattern. Instead, it appears closely associated with structural inequality, differences in expenditure effectiveness, financial protection gaps, and variation in resilience. From a client perspective, the main value of the findings lies in making those differences visible and in showing that future support should be targeted where both current coverage and projected momentum remain comparatively weak.

5. Discussion

The findings suggest that progress toward Universal Health Coverage should be understood as uneven rather than universal. Although global service coverage improved between 2000 and 2022, that progress was not shared evenly across regions, income groups, or countries. This matters because global averages can give a more positive impression than the underlying pattern justifies. Where variation remains wide, overall improvement does not necessarily mean that inequality is narrowing or that weaker-performing systems are catching up (World Health Organization and World Bank, 2025; World Bank, 2023).

One important implication is that UHC should be seen not only as a health-policy goal, but also as a broader development challenge. Persistent differences across regions and income groups suggest that progress is shaped by wider conditions such as fiscal capacity,

institutional strength, service-delivery capability, and administrative effectiveness. This does not mean that lower-resource countries cannot improve, as the country-level results show, but it does suggest that structural conditions remain important in shaping both what is feasible and how quickly change can happen (United Nations, 2015; World Health Organization and World Bank, 2025).

The findings also reinforce the importance of spending, while showing that spending alone is not enough. The positive association between health expenditure and service coverage suggests that investment matters, but the variation around that relationship shows that countries do not translate resources into outcomes in the same way. This means that the policy issue is not only how much countries spend, but how effectively resources are allocated and converted into stronger and more equitable service expansion (World Bank, 2022; Kruk et al., 2018).

Financial protection adds an important layer to this interpretation. Expanding service coverage without reducing the financial burden on households may represent partial progress, but not complete progress. UHC remains limited where people can technically access services but still face catastrophic out-of-pocket costs. This supports a multidimensional interpretation of progress and suggests that reforms focused only on service expansion are unlikely to be sufficient unless they are matched by measures that reduce direct financial pressure on patients (United Nations Statistics Division, 2025b; Wagstaff et al., 2018).

The COVID-period analysis also highlights the importance of resilience. Systems that improve during stable periods but struggle to maintain essential services under pressure look different from systems that are able to absorb disruption more effectively. In that sense, resilience should be treated as part of UHC performance rather than as a separate issue. The period around the pandemic therefore adds an important reminder that progress needs to be durable as well as upward (Kruk et al., 2018; World Health Organization and World Bank, 2025).

The indicative 2030 projection strengthens this point by drawing attention to momentum. Historical progress matters not only as a record of past improvement, but also as a signal of likely future direction if recent long-term patterns continue. These estimates are not forecasts, but they help distinguish between contexts where progress appears relatively stronger and those where it may remain insufficient relative to the time left before 2030 (United Nations, 2015; World Health Organization and World Bank, 2025).

From a Business Analysis and Consulting perspective, the report highlights a problem of prioritisation under uncertainty. Policy support and international attention need to reflect actual variation in performance, affordability, resilience, and likely future trajectory. This is where comparative analysis adds value: it does not suggest that one policy approach fits all countries, but it helps identify where different types of support may be most needed (World Health Organization and World Bank, 2025; World Bank, 2023).

The report also has several limitations. Not all indicators were equally complete across countries and years in the merged workbook, so some analyses were based on broader samples while others required more filtered subsets. The use of internationally comparable secondary data provides breadth, but it does not allow deep causal explanation. The relationship between expenditure and coverage should therefore be interpreted as associative rather than causal. Financial protection data were also less complete than service-coverage

data, which limited the depth of some comparisons. In addition, the 2030 estimates are indicative rather than predictive, because future trajectories may change in response to policy reform, fiscal pressure, political change, or further shocks (Kruk et al., 2018).

Even with these limitations, the report remains useful because it shows where progress appears stronger, where it remains weaker, and why that matters for the years leading to 2030.

6. Conclusion and Recommendations

This report examined progress toward Universal Health Coverage under SDG 3.8 between 2000 and 2022 using three linked indicators: service coverage, financial protection, and health expenditure per capita (United Nations Statistics Division, 2025a; United Nations Statistics Division, 2025b; World Bank, 2022). The analysis showed that progress has taken place, but it has not been evenly shared. Regional and income-group differences remain substantial, country trajectories vary widely, and the COVID-period exposed important differences in resilience. The evidence also indicates that health expenditure is positively associated with service coverage, but not in a simple or uniform way. This suggests that the effectiveness of spending matters alongside the amount of spending itself.

A central conclusion is that global UHC progress cannot be understood through a single average pathway. Some countries and groups appear to be moving along relatively stronger trajectories, while others continue to face structural barriers that weaken both current outcomes and future momentum. Financial protection also remains especially important because access alone does not represent complete progress if households remain vulnerable to catastrophic expenditure (United Nations Statistics Division, 2025b; Wagstaff et al., 2018). The analysis therefore supports a multidimensional view of UHC in which access, affordability, and resilience need to be considered together.

The 2030 perspective strengthens this conclusion. The simple projection used in the report suggests that current trajectories are not equally strong across countries and groups. While these estimates are indicative rather than definitive, they highlight an important practical reality: if recent patterns continue, some countries are likely to remain comparatively behind unless more targeted action is taken. The question is therefore not only whether progress is happening, but whether it is happening quickly enough, broadly enough, and sustainably enough to matter by 2030 (United Nations, 2015).

The report leads to four main recommendations, presented in priority order:

Priority 1: Prioritise support toward countries and groups with weaker projected trajectories. Because progress remains uneven, policy attention should be directed more sharply toward countries and groups where both current performance and long-term momentum remain weak. Comparative evidence from this report suggests that resource allocation, technical support, and policy attention are likely to be more effective when they are targeted toward lower-performing regions, lower-income groups, and countries showing weaker long-term momentum.

Priority 2: Treat financial protection as a core policy priority rather than a secondary issue. The findings show that service coverage alone does not provide a complete picture of UHC progress. Health-system strategies should therefore address affordability directly through

stronger financial-risk protection, lower out-of-pocket burden, and closer alignment between service expansion and household protection.

Priority 3: Focus on spending effectiveness as well as spending volume.

The positive association between expenditure and service coverage suggests that investment matters, but the variation around this relationship indicates that expenditure alone is not enough. Decision-makers should place greater emphasis on how resources are allocated, managed, and converted into service outcomes, particularly in systems where higher spending has not yet produced proportionate improvements.

Priority 4: Build resilience into UHC strategy for the period to 2030.

The COVID-period findings show that progress remains vulnerable to disruption. Health-system resilience should therefore be treated as part of UHC planning rather than as a separate concern. This includes strengthening continuity capacity, maintaining essential services during stress, and reducing the risk that future shocks reverse hard-won gains.

The report also has limitations that should guide future work. The analysis is based on international secondary data, which provide breadth but do not allow deep causal explanation. Some indicators were more complete than others across countries and years, and the 2030 estimates are indicative rather than predictive. Future analysis could therefore build on this work by incorporating more detailed country-level variables such as governance quality, workforce capacity, financing design, or service-delivery structure, and by adding more focused case comparison of stronger and weaker performers.

Overall, the report provides a more decision-relevant view of UHC progress by showing not only that progress has occurred, but where it appears strongest, where it remains weakest, and where targeted support is likely to matter most before 2030.

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8. Appendices

Appendix A: Data Sources and Indicator Definitions

This report uses three main datasets to assess progress toward Universal Health Coverage (UHC) under SDG 3.8. These datasets were selected because they capture the three core dimensions used throughout the analysis: service coverage, financial protection, and health-system investment. This selection is aligned with the project brief, the project plan, and the methodological direction set out in the kickoff guidance, which emphasised trend analysis, regional and income-group comparison, and the relationship between health spending and UHC progress.

1. SDG Indicator 3.8.1 – UHC Service Coverage Index

This indicator measures coverage of essential health services using a composite index reported on a scale from 0 to 100. The index is based on tracer interventions covering reproductive, maternal, newborn and child health, infectious diseases, non-communicable diseases, and service capacity and access. In this report, SDG 3.8.1 is used as the main outcome indicator because it provides the most direct internationally comparable measure of progress in access to essential health services.

2. SDG Indicator 3.8.2 – Financial Protection

This indicator captures the proportion of the population experiencing large household out-of-pocket health expenditures as a share of total household expenditure or income. It is included because access to care alone does not represent full UHC progress if households remain financially vulnerable when using healthcare. In the global monitoring framework, indicators 3.8.1 and 3.8.2 are intended to be interpreted jointly.

3. Health Expenditure per Capita

This variable represents health spending per person and is used in the report as a proxy for health-system investment. It is included to explore whether higher levels of spending are associated with stronger UHC service coverage outcomes. In the analysis, this measure is treated as an associational explanatory variable rather than as proof of causation.

Main data sources used

- United Nations SDG Global Database
- World Health Organization Global Health Observatory
- World Bank Data / World Development Indicators
- Our World in Data

These are the same core sources identified in the kickoff material and project plan.

Appendix B: Data Preparation and Integration

Before analysis, the datasets were cleaned, standardised, and merged into a single country-year dataset. This stage was necessary because the source datasets used slightly different country naming conventions, varied in time coverage, and required additional classification fields for regional and income-group comparison. The data-preparation process followed the same logic described in the project draft and report methodology.

The main preparation steps were as follows:

1. Selection of variables

Only the variables required for analysis were retained from each source dataset. These included country identifiers, year, SDG 3.8.1, SDG 3.8.2, and health expenditure per capita.

2. **Standardisation of country names**

A country-matching process was used to resolve differences in naming conventions across sources. This enabled consistent merging across WHO, UN, and World Bank classification data.

This process was supported through workbook mapping sheets, including country-name harmonisation and identifier standardisation tables.

3. **Assignment of region and income group**

Countries were classified using World Bank economy names, regional groups, and income groups so that comparisons could be made across both geographic and economic dimensions.

4. **Time alignment**

The analysis period was restricted to **2000–2022**. This period was selected because 2022 represented the most recent year for which the key indicators were sufficiently comparable across countries and sources. Later observations were more incomplete and therefore less suitable for consistent cross-country comparison. This was also consistent with the project draft and the methodological rationale used in the report.

5. **Handling of missing values**

Missing observations were retained in the merged dataset where possible, rather than deleting rows immediately. This allowed broad trend analysis to remain intact while enabling more specific filtered analysis for selected tasks such as cross-sectional comparison and correlation.

6. **Creation of merged analytical dataset**

The final merged dataset, referred to as **Master_Merged**, includes:

- Country
- Harmonised World Bank country name
- Year
- Region
- Income group
- UHC service coverage (SDG 3.8.1)
- Financial protection (SDG 3.8.2)
- Health expenditure per capita

A	B	C	D	E	F	G	H	I	J	K	L	M
UN_Code	Country	WB_Name	Year	Region	Income_Group	UHC_3_8_1	Financial_Protection_3_8_2	Health_Expenditure_per_capita	UHC_vis	FP_vis	Spend_vis	Performance_Category
20	Andorra	Andorra	2022	Europe & Central Asia	High income	74.00	4.36	3190.11	74	4.36	3190.11	Emerging Performer
50	Bangladesh	Bangladesh	2022	South Asia	Lower middle income	54.00	41.7	56.62	54	41.7	56.62	Low Performer
64	Bhutan	Bhutan	2022	South Asia	Lower middle income	69.00	8.98	138.01	69	8.98	138.01	Emerging Performer
70	Bosnia and Herzegovina	Bosnia and Herzegovina	2022	Europe & Central Asia	Upper middle income	64.00	8.58	667.3	64	8.58	667.3	Emerging Performer
100	Bulgaria	Bulgaria	2022	Europe & Central Asia	High income	73.00	19.29	1066.69	73	19.29	1066.69	Emerging Performer
120	Cameroon	Cameroon	2022	Sub-Saharan Africa	Lower middle income	47.00	38.84	72.13	47	38.84	72.13	Low Performer
203	Czechia	Czechia	2022	Europe & Central Asia	High income	83.00	11.34	2431.08	83	11.34	2431.08	High Performer
268	Georgia	Georgia	2022	Europe & Central Asia	Upper middle income	71.00	31.75	546.23	71	31.75	546.23	Emerging Performer
300	Greece	Greece	2022	Europe & Central Asia	High income	78.00	14.05	1768.11	78	14.05	1768.11	High Performer
356	India	India	2022	South Asia	Lower middle income	68.00	30.94	81.51	68	30.94	81.51	Emerging Performer
360	Indonesia	Indonesia	2022	East Asia & Pacific	Upper middle income	68.00	27.55	127.44	68	27.55	127.44	Emerging Performer
364	Iran (Islamic Republic of)	Iran, Islamic Rep.	2022	Middle East, North Africa	Upper middle income	81.00	15.65	250.23	81	15.65	250.23	High Performer
392	Japan	Japan	2022	East Asia & Pacific	High income	85.00	10.8	4201.79	85	10.8	4201.79	High Performer
417	Kyrgyzstan	Kyrgyz Republic	2022	Europe & Central Asia	Lower middle income	73.00	9.64	88.08	73	9.64	88.08	Emerging Performer
442	Luxembourg	Luxembourg	2022	Europe & Central Asia	High income	83.00	17.19	7022.63	83	17.19	7022.63	High Performer
458	Malaysia	Malaysia	2022	East Asia & Pacific	Upper middle income	80.00	17.82	458.22	80	17.82	458.22	High Performer
466	Mali	Mali	2022	Sub-Saharan Africa	Low income	41.00	40.07	31.09	41	40.07	31.09	Low Performer
484	Mexico	Mexico	2022	Latin America & Caribbean	Upper middle income	79.00	15.91	650.97	79	15.91	650.97	High Performer
496	Mongolia	Mongolia	2022	East Asia & Pacific	Upper middle income	71.00	27.75	448.24	71	27.75	448.24	Emerging Performer
524	Nepal	Nepal	2022	South Asia	Lower middle income	56.00	24.94	87.34	56	24.94	87.34	Low Performer
566	Nigeria	Nigeria	2022	Sub-Saharan Africa	Lower middle income	47.00	47.35	86.06	47	47.35	86.06	Low Performer
807	North Macedonia	North Macedonia	2022	Europe & Central Asia	Upper middle income	68.00	11.15	561.69	68	11.15	561.69	Emerging Performer
604	Peru	Peru	2022	Latin America & Caribbean	Upper middle income	69.00	28.51	445.1	69	28.51	445.1	Emerging Performer
620	Portugal	Portugal	2022	Europe & Central Asia	High income	84.00	14.58	2600.57	84	14.58	2600.57	High Performer
498	Republic of Moldova	Moldova	2022	Europe & Central Asia	Upper middle income	71.00	10.39	394.42	71	10.39	394.42	Emerging Performer
688	Serbia	Serbia	2022	Europe & Central Asia	Upper middle income	72.00	11.02	843.58	72	11.02	843.58	Emerging Performer

Appendix Figure A1. Example structure of the Master_Merged dataset

This integrated structure provided the analytical basis for trend analysis, regional and income-group comparison, spending-coverage analysis, COVID-period interpretation, and indicative 2030 projection.

Appendix C: Excel Workbook Structure

The analysis was conducted using a structured Excel workbook designed to organise the source data, prepare the merged dataset, and generate analysis outputs for the final report. The workbook combines raw or cleaned source sheets, mapping sheets, merged data, analytical summaries, and report-support visuals.

Core source and preparation sheets

- **UHC_381_clean** – cleaned service coverage dataset for SDG 3.8.1
- **FP_382_Final** – financial protection dataset for SDG 3.8.2
- **Spend_WHO_clean** – health expenditure per capita dataset
- **WB_Class** – World Bank country, region, and income-group classification sheet
- **Country_Name_Fix** – country-name harmonisation table used to align source country names with World Bank economy names
- **Country_Map** – supporting mapping sheet used to standardise country identifiers across datasets
- **Master_Merged** – merged country-year dataset used as the main analysis base

Main analysis sheets

- **Pivot_UHC_Trend** – long-term trend in UHC service coverage
- **Pivot_Region_UHC** – comparison of average UHC service coverage by region
- **Pivot_Income_UHC** – comparison of average UHC service coverage by World Bank income group
- **Scatter_2022** – cross-sectional comparison of health expenditure per capita and UHC service coverage
- **Correlation_Analysis** – summary of the spending-coverage relationship
- **CovidShock_Region** – descriptive comparison of COVID-period UHC change by region
- **CovidShock_IncomeGroup** – descriptive comparison of COVID-period UHC change by income group
- **Country_Progress_Summary** – country-level summary sheet including historical change, average annual change, and projected UHC values to 2030
- **Analysis_Outputs** – supporting summary outputs used for interpretation and cross-checking
- **Table_4_4** – compact summary table used to support the Findings section

The workbook was used as an analytical support tool rather than as an end product in itself. Its main role was to ensure consistent comparison across countries, regions, and income groups while providing a structured basis for the figures and tables reported in the main text.

Appendix D: Notes on Analytical Interpretation and Limitations

Several interpretation points are important for understanding the analysis correctly.

1. Spending and coverage relationship

The comparison between health expenditure per capita and UHC service coverage should be interpreted as an association, not a causal test. The dataset does not include the full range of variables needed to explain national performance in a causal way, such as governance quality, workforce strength, insurance design, or detailed policy implementation.

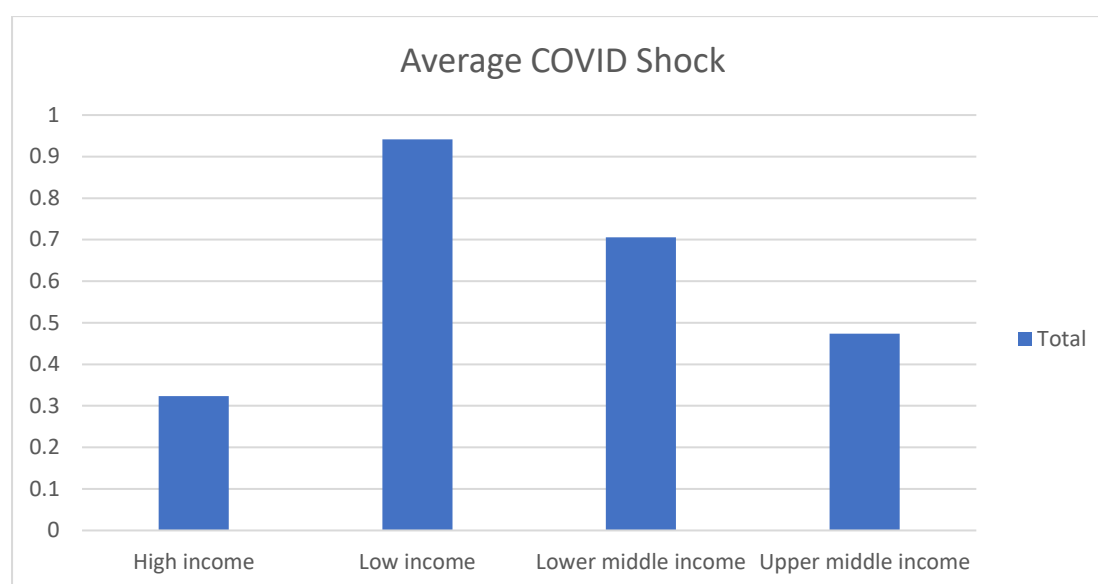
2. Financial protection

Financial protection is included as a core UHC dimension because service expansion alone is not sufficient if households remain exposed to catastrophic out-of-pocket spending. For this reason, the report treats access and affordability as complementary dimensions of progress rather than substitutes.

In this report, this dimension is interpreted cautiously because coverage is more limited across countries and years than for the service-coverage indicator.

3. COVID-period interpretation

The COVID-related analysis in the workbook is descriptive.



Appendix Figure A2. Descriptive COVID-period change in UHC service coverage by income group, 2019–2021

It is intended to show whether recent progress patterns appear to have weakened, held, or recovered across groups. It does not estimate the causal effect of the pandemic in a formal econometric sense. A simple COVID-period change measure was used to compare UHC values between 2019 and 2021:

$$\text{COVID-period change} = \text{UHC 2021} - \text{UHC 2019}$$

4. Indicative 2030 projection

The workbook includes a projected UHC value for 2030 based on a simple continuation of each country's historical average annual change between 2000 and 2022. Annual Change 2000–2022 = $(\text{UHC 2022} - \text{UHC 2000}) / 22$.

This was used to support interpretation of future trajectory, not to generate a definitive forecast. The projected value was calculated as:

$$\text{Projected UHC 2030} = \text{UHC 2022} + (\text{Annual Change 2000–2022} \times 8)$$

This means the 2030 values should be interpreted as **indicative projections**, useful for comparison but not as certain predictions of SDG target achievement.

5. Benchmark-based status categories

Where the workbook groups projected values into broad status categories, these should be treated as analytical support tools rather than official SDG classifications. The report therefore places greater emphasis on projected trajectory and comparative momentum than on any single threshold-based label.